

7(a)	92 protons and 146 neutrons (in nucleus)	B1
	92 (orbital) electrons	B1
7(b)	charge = $2e$ $(= 2 \times 1.60 \times 10^{-19} \text{ C})$	C1
	mass = $4u$ $(= 4 \times 1.66 \times 10^{-27} \text{ kg})$	C1
	ratio = $(2 \times 1.60 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ $= 4.8 \times 10^7 \text{ C kg}^{-1}$	A1
7(c)(i)	up down down / udd	B1
7(c)(ii)	up up up / uuu	B1